

Lattices, T-Duality, and Double Field Theory

on $K3 \times T^2$:

A Lean 4 Companion Formalization

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Abstract

Closed-string compactification on $K3 \times T^2$ carries, besides its continuous geometric moduli, a discrete duality structure: T-duality identifies backgrounds related by $R \leftrightarrow \alpha'/R$ and, more generally, by the arithmetic group $O(d, d; \mathbb{Z})$ acting on the lattice of momentum and winding charges. We report a Lean 4 library, `DualScaleStream2` (19 modules, built on Mathlib, toolchain v4.33.1), that kernel-checks the algebraic skeleton underlying this structure: positive-definiteness and unimodularity of the E_8 and hyperbolic lattices that build $H^2(K3, \mathbb{Z})$; the signature arithmetic of the K3, Mukai, and $K3 \times T^2$ charge lattices; the $O(d, d; \mathbb{Z})$ generators of Givoeon–Porrati–Rabinovici and their charge-norm and spectrum invariance; the Hull–Zwiebach generalized metric of Double Field Theory and its B -field shifts and section condition; a d -dimensional trace-bound generalization, $\text{tr } G + \text{tr } G^{-1} \geq 2d$, of the one-dimensional statement behind this research programme’s “dual-scale” proposal; flux integrality and D-brane tadpole counting on $K3 \times T^2$ and $K3 \times K3$; and the arithmetic of the Eguchi–Ooguri–Tachikawa Mathieu-moonshine coefficients. All 99 theorems are accepted by the Lean 4 kernel with zero `sorry` and an axiom footprint limited to Lean’s three standard axioms (`propext`, `Classical.choice`, `Quot.sound`). We state throughout, and repeat in a dedicated section, what this formalization does and does not establish: every statement is about finite-dimensional matrices over $\mathbb{Z}$ or $\mathbb{R}$; the identification of these matrices with string-theory charge lattices, dualities, and background fields is quoted from the literature (Tier L) or is this project’s own reading (Tier C), never re-derived as physics. We also report, as part of the methods, the tiered multi-agent pipeline that produced these proofs and its measured failure modes: a cheap local prover closes concrete goals and nothing symbolic, and two agent self-reports of successful compilation were false and caught only because every claim was independently recompiled.

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1 Introduction

1.1 The physical question

A closed string propagating on a circle of radius R has momentum modes $n \in \mathbb{Z}$ and winding modes $w \in \mathbb{Z}$, with mass spectrum invariant under the simultaneous exchange $R \leftrightarrow \alpha'/R$, $n \leftrightarrow w$: the Buscher T-duality [2, 3]. Two geometrically distinct backgrounds — a large circle and a small one — describe the same conformal field theory. On a d -torus this discrete symmetry grows into the arithmetic group $O(d, d; \mathbb{Z})$ acting on the $2d$ -dimensional lattice of momenta and windings, preserving the split-signature pairing that couples the two [4, 5]. Compactifying further on a $K3$ surface adds a second, richer duality structure: the moduli space of $(0, 4)$ or $(4, 4)$ superconformal field theories on $K3$ is (conjecturally, and in restricted regimes provably) a double coset of the orthogonal group of an even lattice of signature $(3, 19)$, and type IIA/IIB and heterotic strings on $K3 \times T^2$ or $K3 \times T^3$ are related by a web of string-string and string-M-theory dualities in which T -duality, S -duality, and geometric monodromies of $K3$ all appear as elements of a single arithmetic group [6]. Hull and Zwiebach’s Double Field Theory (DFT) makes the $O(d, d)$ structure manifest at the level of the low-energy effective action, by doubling the torus coordinates and packaging the metric G and Kalb–Ramond field B into a single *generalized metric* $\mathcal{H}(G, B)$ valued in the coset $O(d, d)/(O(d) \times O(d))$ [7].

This research programme’s own recurring theme — explored in Stream 1’s Lean corpus and its companion paper 7 [13] — is that the self-dual point of this duality, $R = \sqrt{\alpha'}$, marks a minimum of an effective length scale $R_{\text{eff}} = R + \alpha'/R$, suggestively read as a stringy obstruction to reaching arbitrarily short distances. The present paper reports a second, independent Lean 4 library, `DualScaleStream2`, that (a) builds the lattice-theoretic scaffolding of $K3 \times T^2$ compactification from first principles (E_8 , the hyperbolic plane U , their direct sums), (b) formalizes the $O(d, d; \mathbb{Z})$ duality group and the Hull–Zwiebach generalized metric for *every* d , not just $d = 1$, and (c) proves a d -dimensional generalization of the dual-scale bound, $\text{tr } G + \text{tr } G^{-1} \geq 2d$, that specializes exactly to Stream 1’s circle statement. It also revisits, with a corrected and more carefully sourced arithmetic, the Mathieu-moonshine decomposition of the $K3$ elliptic genus and the flux/tadpole counting of $K3 \times T^2$ orientifolds.

1.2 What a machine-checked formalization adds, and what it does not

A Lean 4 kernel proof adds a very specific and narrow guarantee: that a stated fact about matrices, vectors, and rational or real arithmetic follows from the axioms of Lean’s core logic plus (in this library) Mathlib’s development of linear algebra, by an explicit, machine-verified derivation with no unproven step (`sorry`) and no undisclosed extra axiom. It does not, by itself, establish that the matrices in question *are* the charge lattices, dualities, or background fields of any string theory — that identification is supplied by the physics literature we cite, and is stated here as such throughout. Nor does formalizing an algebraic identity establish the deeper facts that make it interesting: that $O(d, d; \mathbb{Z})$ is generated by the elements we exhibit (Giveon–Porrati–Rabinovici prove this; we do not re-derive it), that Sylvester’s law of inertia correctly reads off a lattice’s signature from a diagonal congruence (we use it, we do not reprove it), or that the $K3$ elliptic genus really carries an M_{24} action (Eguchi–Ooguri–Tachikawa’s observation, proved by Gannon [12]). We adopt the same three-tier epistemic convention as paper 7, restated below for a self-contained reading, and follow it strictly: every claim in this paper is labelled, and a claim never rests, even transitively, on a lower tier without that dependency being stated.

1.3 Epistemic tiers

- **Tier A — Lean kernel** — **Lean kernel**: the exact stated declaration is accepted by the Lean 4 kernel (v4.33.1, Mathlib v4.33.1), with zero `sorry`, and `#print axioms` shows only `propext`, `Classical.choice`, and `Quot.sound` (Appendix A). Tier A is a claim about the

cited Lean declaration, not automatically about the informal sentence next to it: where a theorem proves only one direction of an “iff”, or a bound over a strict subset of cases, this paper says so.

- **Tier L — literature** — **literature**: quoted from a cited paper, by section/equation and by file/line into the `papers/foundations/*.txt` extraction used to write this corpus, not independently re-derived here.
- **Tier C — conjecture** — **conjecture**: this project’s own interpretation, physical reading, or unproven proposal.

A fourth marker, `symbolic check (sympy)`, not Tier A, flags a small number of matrix-convention checks (sign and transpose placements) that were verified with `sympy` *before* the Lean statement was written, as a sanity check on the formalization choice, and are explicitly *not* Tier A: a symbolic computer-algebra check is not a kernel proof, and this paper never cites one as if it were.

Forbidden phrasings. Following `docs/PAPER_REVISION_BRIEF_2026_09_17.md` §0, this paper does not use the phrases “zero axioms,” “100% mathematical rigor,” “certifies stability,” “formalizes string theory,” “first fully certified,” or “all of physics.” The correct form used throughout is: proved in Lean 4 with no axioms beyond Lean’s three standard axioms, with the string-theoretic reading of each proved statement stated separately and tiered.

1.4 Relation to the Stream 1 corpus: why this library uses Mathlib

Paper 7’s corpus (`StringTheoryFoundation`, `DoubleFieldTheory`, `DualScaleM24Formalization`, `DualScaleValidation`, `Lean5Corpus`, `StringTheoryFormalization`) deliberately has *no* Mathlib dependency, so every statement there is over `Nat`, `Int`, or an ad hoc rational-pair structure; its circle-bound theorem, `UseCase1.self_dual_is_global_minimum`, is consequently only an $\mathbb{N}$ -instance ($R \geq 1 \Rightarrow R^2 + 1 \geq 2$), not a real bound. `DualScaleStream2` is a new, separate `lean_lib` that *imports* Stream 1 (never the reverse, so Stream 1’s build cannot be broken by Stream 2 work) and additionally imports Mathlib, specifically to reach real-number linear algebra: positive-definite matrices, matrix inverses, traces, and transposes. This is the direct, disclosed reason Section 5’s bound is a genuine statement over $\mathbb{R}$ for every positive-definite $d \times d$ matrix G (`DualScale.dualScale_ge`, `DualScale.circle_effective_scale_ge_two`), where paper 7’s analogous statement remained an integer instance pending exactly this step. We regard this as the two corpora being complementary rather than in tension: Stream 1 demonstrates what can be proved with *no* external dependency; Stream 2 demonstrates what the same programme’s central claims look like once Mathlib is available.

The remainder of this paper proceeds one formalized layer at a time (lattices, Section 2; T-duality, Section 3; Double Field Theory, Section 4; the dual-scale bound, Section 5; flux and tadpole, Section 6; Mathieu moonshine, Section 7), each with the ordinary-notation mathematics, its physical motivation, and a table mapping every cited Lean declaration to its tier. Section 8 then collects, in one place, an explicit statement of what is and is not proved. Section 9 describes the tiered multi-agent pipeline that produced the corpus and its measured results, including its failures. Appendix A gives exact reproduction commands and numbers.

Throughout, charge vectors are ordered momentum $\oplus$ winding, and the $O(d, d)$ -invariant form is $\eta = \begin{pmatrix} 0 & \mathbf{1}_d \\ \mathbf{1}_d & 0 \end{pmatrix}$.

2 Lattices for $K3 \times T^2$

2.1 Why lattices

The charge lattices of string compactification — H^2 of a $K3$ surface, the Narain lattice of a torus, the lattice of D-brane charges — are even, unimodular (self-dual) lattices of indefinite signature, and their isometry groups are exactly the duality groups of the physical theory. Two building blocks recur throughout the subject: the negative-definite E_8 root lattice $E_8(-1)$, and the rank-2 hyperbolic plane U with Gram matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Huybrechts records the resulting decomposition of the $K3$ lattice,

$$H^2(K3, \mathbb{Z}) \cong E_8(-1)^{\oplus 2} \oplus U^{\oplus 3}, \quad (1)$$

an even, unimodular lattice of rank 22 and signature (3, 19) (Huybrechts, *Lectures on K3 Surfaces*, Ch. 1, eq. (3.4), `huybrechts_K3Global.txt` l. 603) [8]. Its orthogonal group, and in particular the subgroup fixing a choice of polarization, is the discrete part of the $K3$ moduli space and of the string dualities acting on it (Aspinwall, *K3 Surfaces and String Duality*, hep-th/9611137, ll. 705–766, 1737–1748) [6].

2.2 E_8 : unimodularity, positive-definiteness, and the exact determinant

E_8 is the rank-8 integral lattice with Cartan/Gram matrix

$$\text{cartanE8} = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 2 \end{pmatrix}. \quad (2)$$

Rather than asserting $\det = \pm 1$, `DualScaleStream2` certifies unimodularity the way `Lattice.Basic.isUnimodular_of_mul_eq_one` makes canonical for the whole corpus: an explicit integer matrix `cartanE8Inv` is exhibited, and the kernel checks `cartanE8Inv * cartanE8 = 1` entry by entry (`cartanE8Inv_mul`); an integer left inverse forces $\det = \pm 1$ by `isUnimodular_of_mul_eq_one`. This certificate-based style means a wrong inverse would simply fail to typecheck rather than silently asserting a false determinant. The matrix is symmetric (`cartanE8_symm`) with even diagonal (`cartanE8_evenDiag`), so by `even_quadratic_form_of_even_diag` every integer vector has even norm — E_8 is an even lattice. The sign-reversed lattice $E_8(-1)$, the summand that actually appears in $H^2(K3, \mathbb{Z})$, inherits all three properties (`e8Neg_symm`, `e8Neg_evenDiag`, `e8Neg_unimodular`).

Unimodularity alone does not distinguish E_8 from an indefinite even unimodular lattice of the same rank; the physically relevant fact is that E_8 is *positive definite* (so $E_8(-1)$ is negative definite, matching its role in the signature-(3, 19) decomposition). This is proved by an exact rational LDL^T factorization over $\mathbb{R}$, $D = \text{diag}\left(2, \frac{3}{2}, \frac{4}{3}, \frac{5}{4}, \frac{6}{5}, \frac{7}{6}, \frac{8}{7}, \frac{1}{8}\right)$, with all eight pivots strictly positive and product 1 (`e8_LDL`, `cartanE8_det` : `cartanE8.det = 1` over $\mathbb{Z}$). Positive-definiteness of the real matrix `cartanE8R` (the $\mathbb{Z} \rightarrow \mathbb{R}$ cast of `cartanE8`) is then proved (`cartanE8_posDef`) via the sum-of-squares identity

$$x^T E_8 x = 2y_0^2 + \frac{3}{2}y_1^2 + \frac{4}{3}y_2^2 + \frac{5}{4}y_3^2 + \frac{6}{5}y_4^2 + \frac{7}{6}y_5^2 + \frac{8}{7}y_6^2 + \frac{1}{8}y_7^2, \quad (3)$$

each y_k an explicit linear form in x from the back-substitution of the factorization, checked by `ring` and then completed by an explicit back-substitution argument that a vanishing sum of these eight squares forces $x = 0$.

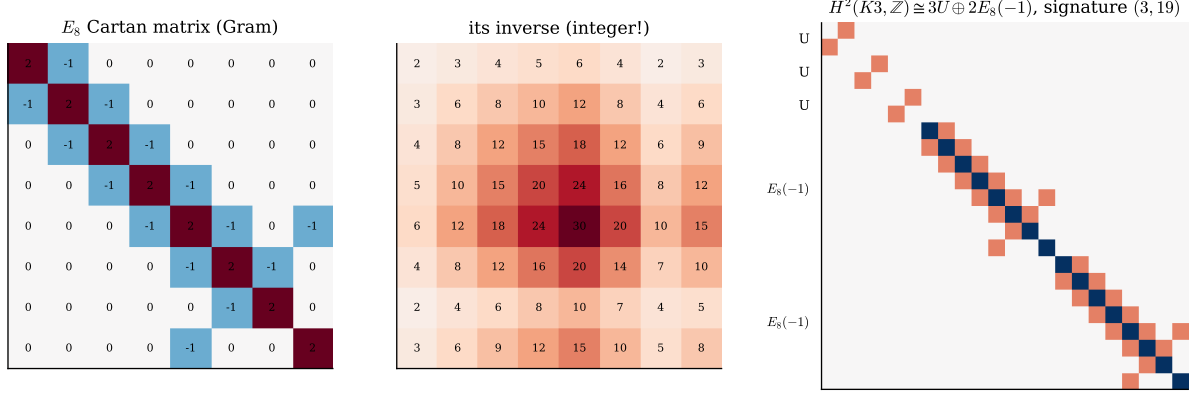


Figure 1: Left: the Gram matrix of E_8 (the Cartan matrix used in Lean) and its inverse, which has integer entries because $\det = 1$ (unimodularity). Right: the 22×22 Gram matrix of the K3 lattice $3U \oplus 2E_8(-1)$ (red +, blue -); even diagonal, determinant ± 1 , signature (3, 19). *Computed figure; data: Lean `cartanE8` (`E8.lean`), `hyperbolicU` (`Hyperbolic.lean`); `cartanE8Inv_mul`, `sigK3_eq`.*

2.3 The hyperbolic plane U

$U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is even (`hyperbolicU_evenDiag`), symmetric (`hyperbolicU_symm`), and its own inverse, hence unimodular (`hyperbolicU_mul_self`, `hyperbolicU_unimodular`). A rational congruence witness $P^T U P = \text{diag}(2, -2)$ with $P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ (`hyperbolicU_congruence`) exhibits one positive and one negative diagonal entry; reading off the lattice’s signature (1, 1) from this diagonal congruence is an instance of Sylvester’s law of inertia, which is [Tier L — literature](#), not re-proved here. Separately, `hyperbolicU_eq_narain_gram` proves that this U is, on the nose, the Gram matrix already certified as the Narain lattice $\Gamma^{1,1}$ in Stream 1’s `StringTheoryFormalization.UseCases.NarainLattice` — Stream 2 builds on that fact rather than restating it.

2.4 Signature bookkeeping: K3, Mukai, and $K3 \times T^2$

`Lattice.K3T2Signature` introduces a small `Signature` structure (a pair of naturals, positive/negative multiplicity) and computes, purely arithmetically, the signatures of

$$\text{sigK3} = E_8(-1) \oplus E_8(-1) \oplus U \oplus U \oplus U, \quad (4)$$

$$\text{sigMukai} = \text{sigK3} \oplus U, \quad (5)$$

$$\text{sigT2} = U \oplus U, \quad (6)$$

$$\text{sigK3T2} = \text{sigMukai} \oplus \text{sigT2}, \quad (7)$$

where the component signatures $U = (1, 1)$ and $E_8(-1) = (0, 8)$ are Tier L inputs (established above only for the individual lattice-theoretic facts unimodular/even/positive-definite, not for the signature label itself, which uses Sylvester’s law). The results, $\text{sigK3} = (3, 19)$, $\text{sigMukai} = (4, 20)$, and $\text{sigK3T2} = (6, 22)$ (`sigK3_eq`, `sigMukai_eq`, `sigK3T2_eq`, each proved by `rfl` on the additive structure), reproduce the Mukai lattice $\Gamma^{4,20}$ and the $K3 \times T^2$ charge lattice $\Gamma^{6,22}$ of the physics literature (Aspinwall, ll. 1737–1748), of ranks 22 and 28 respectively (`rank_K3`, `rank_K3T2`). All three signatures satisfy the index condition $b_+ - b_- \equiv 0 \pmod{8}$ (`index_mod_eight`) required of an even unimodular indefinite lattice by the Milnor–Serre classification (not downloaded; cited by reference only) — again arithmetic on the already-computed pairs, not a re-derivation of the classification theorem itself.

A genuine cross-check, not merely a restatement, is `sigK3_matches_hodge`: the lattice-decomposition signature $(3, 19)$ agrees, component by component, with the *Hodge-theoretic* signature $(2h^{2,0} + 1, h^{1,1} - 1)$ already computed independently in Stream 1’s `StringTheoryFormalization.UseCases.K3Signature`. As the module’s own docstring states, this is agreement between two independently tabulated Tier L inputs (the lattice decomposition and the Hodge diamond), not an independent derivation of either.

2.5 The Mukai pairing

On the extended lattice $H^0 \oplus H^2 \oplus H^4$, Huybrechts’s Mukai pairing on a vector $v = (r, c, s)$ is $\langle v, w \rangle = c \cdot L \cdot c' - rs' - sr'$ (Huybrechts, *Lectures on K3 Surfaces*, Ch. 9 §1, Def. 1.4, chapter heading l. 7352, definition ll. 7475–7482), with Hirzebruch–Riemann–Roch reading $\chi(E, F) = -\langle v(E), v(F) \rangle$. `Lattice.Mukai` represents a Mukai vector as a structure `MukaiVec` ($r : \mathbb{Z}, c : \text{Fin } n \rightarrow \mathbb{Z}, s : \mathbb{Z}$) and proves the pairing `mukaiPair` symmetric for symmetric L (`mukaiPair_symm`), with self-pairing $v^2 = c \cdot L \cdot c - 2rs$ (`mukaiPair_self`), and even whenever L is even (`mukaiPair_even`, reusing `even_quadratic_form_of_even_diag`). The $H^0 \oplus H^4$ summand

$U(-1) = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ is separately shown unimodular (`hyperbolicUNeg_unimodular`). Finally, for *any* Gram matrix L (the statement does not need L even, symmetric, or of any particular rank, because the structure sheaf’s Mukai vector $v = (1, 0, 1)$ has vanishing c -component),

$$\text{mukaiPair}_L((1, 0, 1), (1, 0, 1)) = -2 \quad (\text{structureSheaf_mukai_sq}), \quad (8)$$

which under Riemann–Roch gives $\chi(\mathcal{O}_X, \mathcal{O}_X) = -\langle v, v \rangle = 2$ — but both the identification $v(\mathcal{O}_X) = (1, 0, 1)$ and $\chi = -\langle v, v \rangle$ are Tier L inputs, not proved by this arithmetic identity.

2.6 Reflections and the Weyl group of $E_8(-1)$

For a (-2) -vector v (i.e. $v^\top L v = -2$) in an integer lattice with symmetric Gram matrix L , the reflection $s_v = 1 + vv^\top L$ acting by $x \mapsto x + \langle x, v \rangle v$ is the Weyl reflection used throughout Huybrechts’s discussion of (-2) -curves and Picard–Lefschetz monodromy (Ch. 7 §5.4, ll. 6031–6043; Ch. 6, l. 5400, for the monodromy interpretation) [8]. `Lattice.Reflection` proves, for any rank n and any symmetric integer L , the key matrix identity $vv^\top L vv^\top = (v^\top L v) \cdot vv^\top$ (`vecMulVec_mul_self`, true for any L , no symmetry needed), from which `reflection_isometry` shows that $v^\top L v = -2$ makes s_v an isometry ($s_v^\top L s_v = L$), and `reflection_involution` shows $s_v^2 = 1$. Every simple root e_i of $E_8(-1)$ has norm -2 (`e8Neg_simpleRoot_norm`, checked for each i by `fin_cases`), so each simple Weyl reflection is an isometry of $E_8(-1)$ (`e8Neg_weyl_isometry`). That the simple reflections *generate* the full Weyl group of E_8 is standard and Tier L, not proved here.

3 The T-duality group $O(d, d; \mathbb{Z})$

3.1 Generators

For a d -torus, the charge lattice is $\Gamma^{d,d} \cong \mathbb{Z}^d \oplus \mathbb{Z}^d$ (momentum $\oplus$ winding), with invariant form $\eta = \begin{pmatrix} 0 & I_d \\ I_d & 0 \end{pmatrix}$; $g \in O(d, d; \mathbb{Z})$ means $g^\top \eta g = \eta$. Giveon, Porrati, and Rabinovici’s review (hep-th/9401139, §2.4 “ $O(d, d, \mathbb{Z})$ Duality for d -Dimensional Toroidal Compactifications,” ll. 1355–1373) exhibits two families of generators: the integer antisymmetric Θ -*shift* $g_\Theta = \begin{pmatrix} I & \Theta \\ 0 & I \end{pmatrix}$ (eq. (2.4.25)), and *basis changes* $\text{diag}(A, B)$ with $A \in GL(d, \mathbb{Z})$ acting on the torus lattice. Stream 2 proves, for *every* d (the Lean statements are universally quantified over $d : \mathbb{N}$, not specialized to $d = 1, 2$ as the physics literature usually states them):

Table 1: Lattice layer: Lean declaration and epistemic tier.

Lean declaration	Tier / content
<code>isUnimodular_of_mul_eq_one</code>	A — integer left inverse $\Rightarrow \det = \pm 1$
<code>even_quadratic_form_of_even_diag</code>	A — even diagonal $\Rightarrow$ even quadratic form
<code>cartanE8Inv_mul, cartanE8_symm/_evenDiag/_unimodular</code>	A — E_8 even, symmetric, unimodular
<code>e8Neg_symm/_evenDiag/_unimodular</code>	A — same for $E_8(-1)$
<code>e8_LDL, cartanE8_det</code>	A — exact LDL^T , $\det_{\mathbb{Z}} E_8 = 1$
<code>cartanE8_posDef</code>	A — E_8 positive definite over $\mathbb{R}$
<code>hyperbolicU_symm/_evenDiag/_unimodular/_congruence</code>	A — U even, symmetric, unimodular, signature with n
<code>hyperbolicU_eq_narain_gram</code>	A — $U = \text{Stream 1's Narain Gram matrix}$
$H^2(K3, \mathbb{Z}) \cong E_8(-1)^{\oplus 2} \oplus U^{\oplus 3}$; K3 moduli quotient	L — Huybrechts (3.4); Aspinwall
<code>sigK3_eq/_sigMukai_eq/_sigK3T2_eq, rank_K3/_K3T2, index_mod_eight</code>	A — signature/rank arithmetic
<code>sigK3_matches_hodge</code>	A — cross-check vs. Stream 1's Hodge signature
Mod-8 theorem (Milnor/Serre)	L — not downloaded, cited by reference
<code>mukaiPair_symm/_self/_even</code>	A — Mukai pairing symmetric, self-pairing formula,
<code>hyperbolicUNeg_unimodular</code>	A — $U(-1)$ unimodular
<code>structureSheaf_mukai_sq</code>	A — $v(1, 0, 1)^2 = -2$ for any Gram L
$v(\mathcal{O}_X) = (1, 0, 1)$, $\chi = -\langle v, v \rangle$	L — Huybrechts Ch. 9 Def. 1.4
<code>vecMulVec_mul_self/_reflection_isometry/_involution</code>	A — (-2) -vector reflections are involutive isometries
<code>e8Neg_simpleRoot_norm/_weyl_isometry</code>	A — simple roots of $E_8(-1)$ have norm -2
Weyl group generated by simple reflections	L — standard, not proved here

- `thetaShift_isODD`: if $\Theta^T = -\Theta$ then $g_\Theta \in O(d, d; \mathbb{Z})$ — *only this direction* is proved; the paper does not claim, and the Lean statement does not say, that g_Θ preserving η forces Θ antisymmetric;
- `basisChange_isODD`: $A^T B = 1 \Rightarrow \text{diag}(A, B) \in O(d, d; \mathbb{Z})$;
- `eta_mul_self, eta_isODD`: the full duality η (momentum $\leftrightarrow$ winding) is itself an involution and lies in $O(d, d; \mathbb{Z})$;
- `isODD_mul`: $O(d, d; \mathbb{Z})$ is closed under matrix multiplication.

At $d = 1$, η reindexes exactly to the hyperbolic plane U of Section 2 (`eta_one_reindex`), so the $d = 1$ invariant form and Stream 1's Narain Gram matrix are literally the same matrix, not merely isomorphic ones. That the elements above *generate* all of $O(d, d; \mathbb{Z})$ is stated by GPR (l. 1422) and is Tier L; it is not attempted here.

3.2 Factorized dualities and spectrum equivalence

GPR's eq. (2.4.29) (ll. 1399–1422) exhibits, for each circle k of the torus, a *factorized duality* $D_k = \begin{pmatrix} I - e_k & e_k \\ e_k & I - e_k \end{pmatrix}$, e_k the rank-one projector onto the k -th coordinate, described there as “a generalization of the $R \rightarrow 1/R$ circle duality in the X^i direction.” `TDuality.Factorized` proves each $D_k \in O(d, d; \mathbb{Z})$ (`factorized_isODD`), an involution (`factorized_mul_self`), and that distinct D_k, D_l commute (`factorized_comm`); on T^2 , $D_0 D_1 = \eta$, dualizing both circles recovers the full duality (`factorized_two`). The charge norm $Z^T \eta Z = 2n \cdot w$ for $Z = (n, w)$ (`chargeNorm_sumElim`) is always even, i.e. $\Gamma^{d,d}$ is an even lattice generalizing Stream 1's rank-1 statement (`chargeNorm_even`), and is invariant under every element of $O(d, d; \mathbb{Z})$, not merely the generators (`chargeNorm_invariant`).

A stronger structural fact, *spectrum equivalence* (`TDuality.Spectrum`), goes beyond the generators: every $g \in O(d, d; \mathbb{Z})$ is invertible over $\mathbb{Z}$ with inverse $\eta g^T \eta$, itself in $O(d, d; \mathbb{Z})$ (`isODD_left_inv/_right_inv/_inv_isODD`). Consequently, for *any* real background H and *any* $g \in O(d, d; \mathbb{Z})$, there is an explicit bijection e of the integer charge lattice (namely $Z \mapsto g \cdot Z$, with inverse $Z \mapsto \eta g^T \eta \cdot Z$) under which the mass form and charge norm of the dual background $g^T H g$

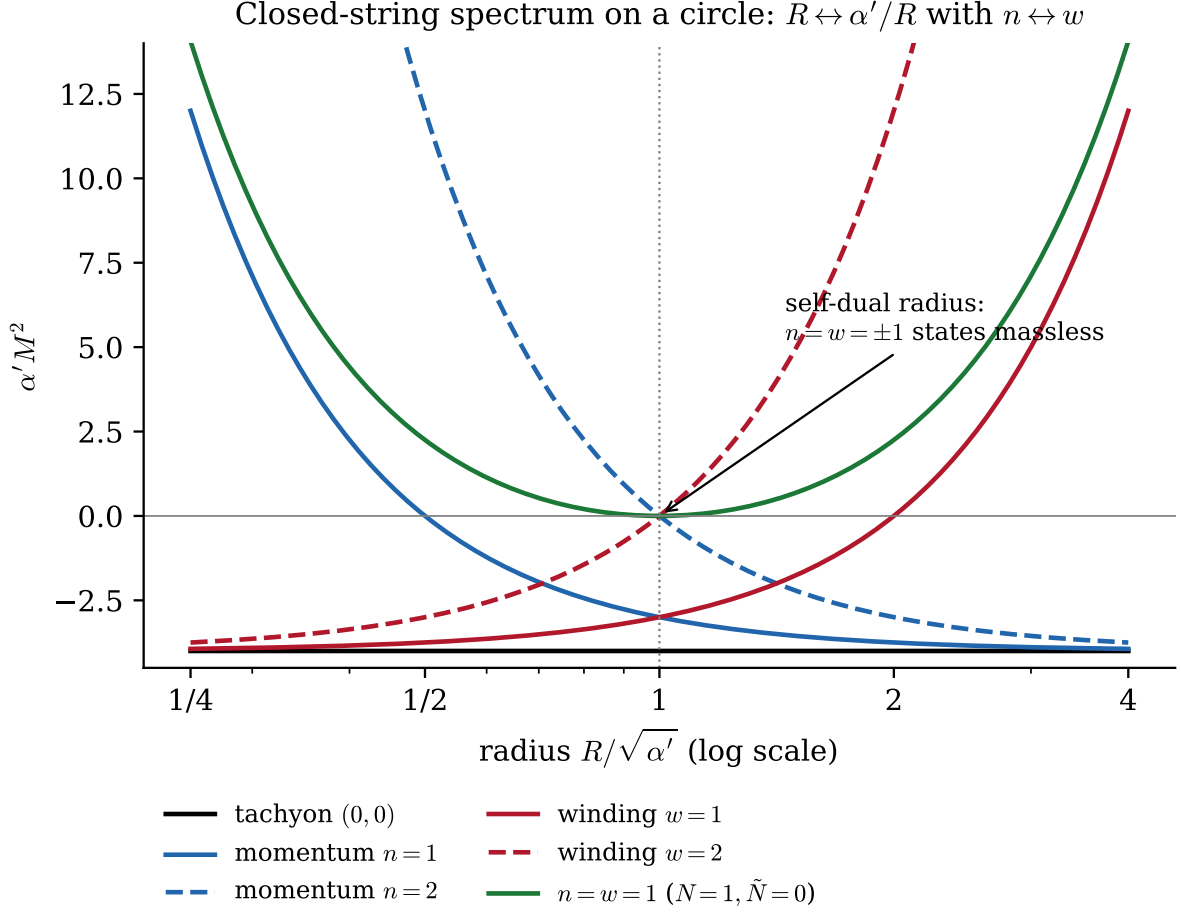


Figure 2: Lowest mass levels of the closed bosonic string on a circle of radius R as a function of $\log R$, with the lowest oscillator levels allowed by level matching $N - \tilde{N} = nw$. Momentum and winding curves are mirror images under $R \mapsto \alpha'/R$; at the self-dual radius the $n = w = \pm 1$ states become massless (enhanced gauge symmetry). *Computed figure; data: Tong §8.2, ll. 11351–11470; Lean `leftMomentum`, `rightMomentum` (units $\alpha' = 1$).*

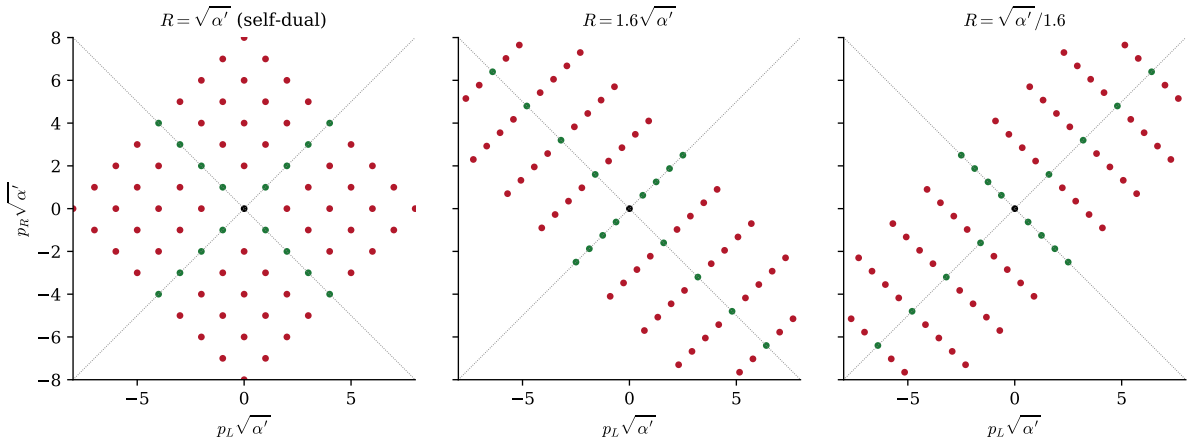


Figure 3: The Narain lattice $\Gamma^{1,1}$ of zero-mode momenta (p_L, p_R) for $|n|, |w| \leq 4$ at three radii. Changing R is a Lorentz boost in the (p_L, p_R) plane that preserves $p_L^2 - p_R^2 = 4nw$ (dotted light-cone lines); the lattices at R and α'/R coincide after $p_R \mapsto -p_R$ — T-duality. *Computed figure; data: Lean `leftMomentum`, `rightMomentum`; GPR §2.3; Tong §8.2.*

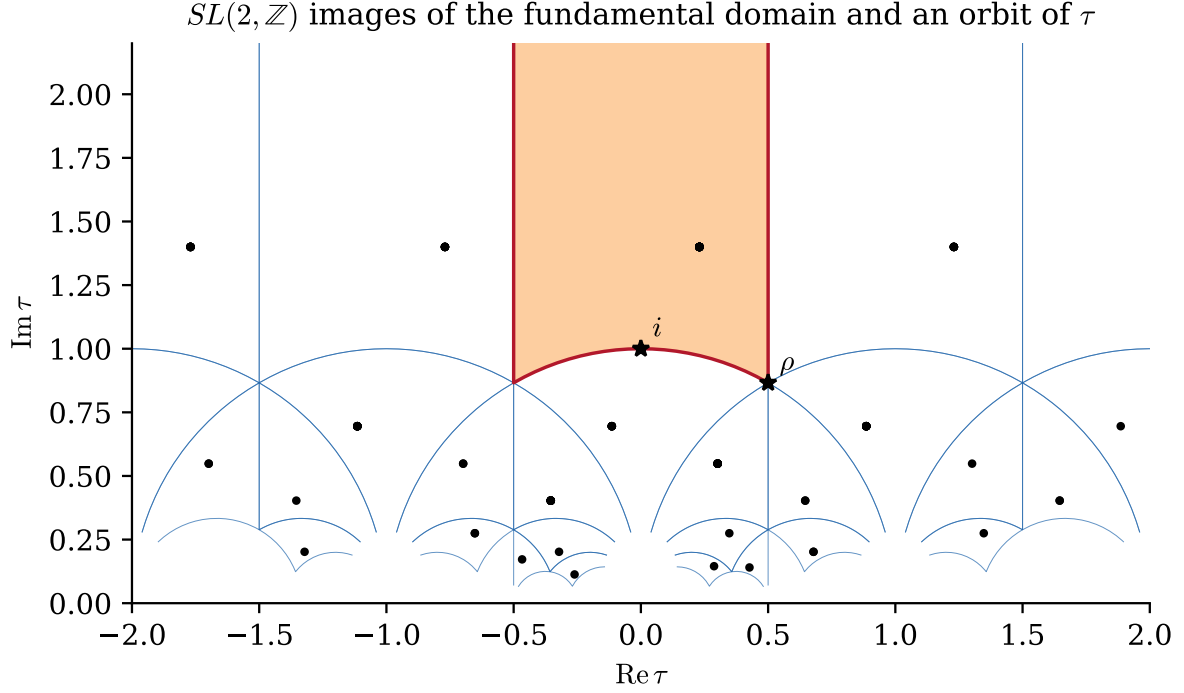


Figure 4: The standard fundamental domain of $SL(2, \mathbb{Z})$ (shaded) and its images under words of length ≤ 4 in $S : \tau \mapsto -1/\tau$ and $T : \tau \mapsto \tau + 1$, together with the orbit of one point. For T^2 the same picture holds for both moduli τ and ρ , exchanged by mirror symmetry. *Computed figure; data: GPR §2.5 / Tong §6 (modular group); Lean `tauShift`, `mirror_conjugates_tauShift`.*

at Z equal those of H at $e(Z)$ (`spectrum_equivalence`). This is a charge-by-charge matching of the discrete (mass², level) data of two $O(d, d; \mathbb{Z})$ -related backgrounds; GPR’s statement that the *full* partition function (with its oscillator sums and modular integral) is invariant (l. 1417) is a strictly stronger claim, Tier L, and is not what is proved here.

3.3 Mirror symmetry on T^2

On T^2 , GPR remark (l. 344) that for a complex torus “mirror symmetry is identical to what is called ‘factorized duality.’” `TDuality.Mirror` verifies the exact integer identity, first checked with `sympy` (`symbolic check (sympy)`, not Tier A),

$$D_0 \cdot \text{diag}(T, (T^T)^{-1}) \cdot D_0 = g_\Theta^T, \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \Theta = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad (9)$$

i.e. the factorized duality conjugates the unimodular basis change T (the $\tau \mapsto \tau + 1$ generator acting on the complex-structure modulus) into the transpose of an antisymmetric Θ -shift (`mirror_conjugates_tauShift`), which by Section 4’s `DFT.BShift.genMetric_bshift` is exactly the form that shifts the generalized metric’s B -field integrally. This is the algebraic content behind “factorized duality exchanges the complex-structure and Kähler moduli of T^2 ”; which *sign* of the induced ρ -shift this corresponds to is a convention-dependent orientation choice for $\rho = B + i\sqrt{\det G}$ and is not claimed.

3.4 $SL(2, \mathbb{Z}) \times SL(2, \mathbb{Z})$ at $d = 2$

GPR’s “The $d = 2$ Example” (ll. 1874–1884) states that the full T^2 duality group is $SL(2, \mathbb{Z})_\tau \times SL(2, \mathbb{Z})_\rho \otimes_S [\mathbb{Z}_2 \times \mathbb{Z}_2]$ — explicitly *not* a naive factorization of $O(2, 2; \mathbb{Z})$. `TDuality.SL2Product`

formalizes a deliberately narrower slice: with $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, the identity $AJA^\top = \det(A) \cdot J$ for every 2×2 integer A (`mul_jMat_mul_transpose`, sympy-checked first, `symbolic check (sympy)`, not Tier A, then proved by `ring` in Lean, so this one *is* Tier A as stated), together with the monoid law for basis changes (`basisChange_mul`) and the additive group law for Θ -shifts (`thetaShift_mul`). The headline result, `basisChange_comm_thetaShift`, proves that a basis change with $\det A = 1$ commutes with every ρ -translation g_{tJ} (an integer Θ -shift along J). **Only this direction is proved**; the converse — that the commutator is nonzero when $\det A \neq 1$ — was checked symbolically with sympy during development and is explicitly *not* a Lean theorem (`symbolic check (sympy)`, not Tier A): it must not be read as Tier A. Consequently what is certified is that $SL(2, \mathbb{Z})_\tau$ and the ρ -translations generate a commuting product inside $O(2, 2; \mathbb{Z})$; the $\rho \mapsto -1/\rho$ generator and the full $SL(2, \mathbb{Z})_\rho$ are not formalized.

Table 2: T-duality layer: Lean declaration and epistemic tier.

Lean declaration	Tier / content
<code>thetaShift_isODD</code>	A — $\Theta^\top = -\Theta \Rightarrow g_\Theta \in O(d, d; \mathbb{Z})$ (one direction)
<code>basisChange_isODD</code>	A — $A^\top B = 1 \Rightarrow \text{diag}(A, B) \in O(d, d; \mathbb{Z})$
<code>eta_mul_self/_isODD, isODD_mul</code>	A — η involutive $\in O(d, d; \mathbb{Z})$; closure
<code>eta_one_reindex</code>	A — $d = 1$: η = Stream 1’s Narain Gram matrix
Generators generate all of $O(d, d; \mathbb{Z})$	L — GPR l. 1422, not proved here
<code>factorized_isODD/_mul_self/_comm/_two</code>	A — $D_k \in O(d, d; \mathbb{Z})$, involutions, commute, $D_0 D_1 = \eta$
<code>chargeNorm_even/_invariant</code>	A — $Z^\top \eta Z$ even, $O(d, d; \mathbb{Z})$ -invariant
<code>isODD_left_inv/_right_inv/_inv_isODD</code>	A — every $g \in O(d, d; \mathbb{Z})$ invertible, inverse $\eta g^\top \eta$
<code>spectrum_equivalence</code>	A — charge-by-charge (mass ² , level) matching under any g
Full partition function invariance	L — GPR l. 1417, strictly stronger, not proved
<code>mirror_conjugates_tauShift</code>	A — exact matrix identity (sympy-checked convention, <code>symbolic check (sympy)</code> , not T
Factorized duality = mirror symmetry	L — GPR l. 344
<code>mul_jMat_mul_transpose</code>	A — $AJA^\top = \det(A)J$
<code>basisChange_comm_thetaShift</code>	A — $\det A = 1 \Rightarrow$ commutes with ρ -translations (one direction)
Converse ($\det A \neq 1 \Rightarrow$ noncommuting)	<code>symbolic check (sympy)</code> , not Tier A — sympy only, not a Lean theorem
Full $SL(2, \mathbb{Z})_\tau \times SL(2, \mathbb{Z})_\rho$	L — GPR ll. 1874–1884, not claimed

4 Double Field Theory

4.1 The generalized metric

Hull and Zwiebach’s Double Field Theory packages the torus metric G and Kalb–Ramond field B into a single object valued in the coset $O(d, d)/(O(d) \times O(d))$, the *generalized metric* (arXiv:0904.4664, eq. (2.17), ll. 662–672)

$$\mathcal{H}(G, B) = \begin{pmatrix} G - BG^{-1}B & BG^{-1} \\ -G^{-1}B & G^{-1} \end{pmatrix}, \quad \mathcal{H}^{-1} = \eta \mathcal{H} \eta. \quad (10)$$

`DFT.GeneralizedMetric` proves, over $\mathbb{R}$ for every d , the constraint in the form $(\eta \mathcal{H})^2 = 1$ (`etaR_genMetric_sq`) requiring *only* that G be invertible — neither symmetry of G nor antisymmetry of B is used in this proof, which is sharper than Hull–Zwiebach’s own statement (stated there for the physical (G, B)). Symmetry of $\mathcal{H}$ under the physical hypotheses $G^\top = G$, $B^\top = -B$ is proved separately (`genMetric_symm`). At $B = 0$, conjugating by the full duality inverts the metric, $\eta \mathcal{H}(G, 0) \eta = \mathcal{H}(G^{-1}, 0)$ (`tduality_inverts_metric`): T-duality *is* the map $G \mapsto G^{-1}$ at the level of the generalized metric. The mass form $Z^\top \mathcal{H} Z$ is covariant under *any* matrix g (again no $O(d, d)$ hypothesis is needed for this identity: `massForm(g^\top H g, Z) = massForm(H, gZ)`, `massForm_covariant`), which is what makes the spectrum-equivalence argument of Section 3

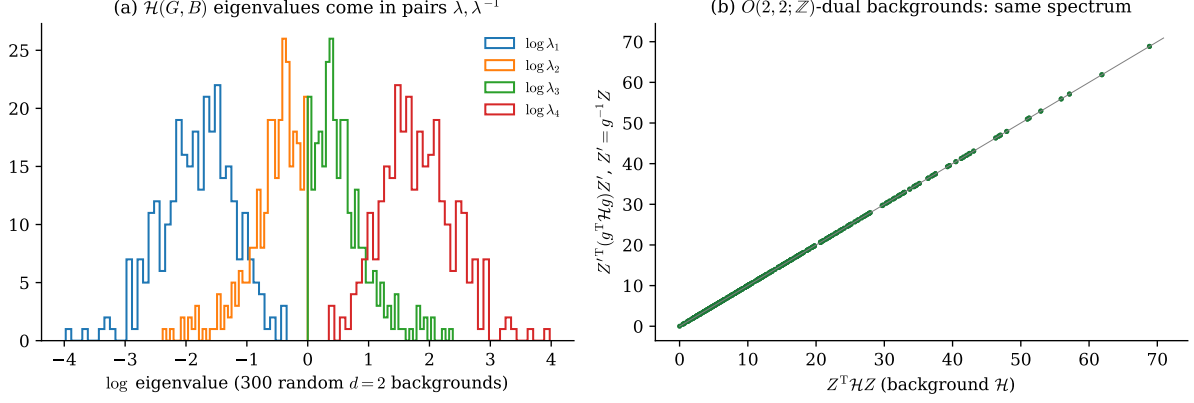


Figure 5: (a) Logarithms of the four eigenvalues of the generalized metric $\mathcal{H}(G, B)$ for 300 random backgrounds on T^2 : they are symmetric about 0, because $(\eta\mathcal{H})^2 = 1$ (`etaR_genMetric_sq`) forces eigenvalues in pairs λ, λ^{-1} . (b) For $g = \Theta \cdot A \cdot (\text{factorized duality}) \cdot \Theta' \in O(2, 2; \mathbb{Z})$, the mass form of every charge Z in background $\mathcal{H}$ equals that of $g^{-1}Z$ in the dual background $g^T \mathcal{H} g$: the points lie on the diagonal (`spectrum_equivalence`, `massForm_covariant`). *Computed figure; data: Lean `genMetric` (`GeneralizedMetric.lean`), `thetaShift`, `basisChange`, `factorized`; HHZ §2.*

work once $g \in O(d, d; \mathbb{Z})$ is assumed there. On a circle ($d = 1, G = R^2, B = 0$), `massForm_circle` proves the DFT mass form equals exactly one half of Stream 1’s $P_L^2 + P_R^2 = \text{momentumMassSq}$ — so Stream 2’s general- d DFT layer and Stream 1’s original circle computation are proved identical at $d = 1$, not merely analogous.

4.2 B -field shifts

GPR’s remark that the Θ -shift “gives only topological contributions” to the spectrum (eq. (2.4.25) context) is realized at the level of the generalized metric: for invertible symmetric G and antisymmetric B, Θ ,

$$g_\Theta \cdot \mathcal{H}(G, B) \cdot g_\Theta^T = \mathcal{H}(G, B + \Theta) \quad (\text{genMetric_bshift}), \quad (11)$$

so the periodic identification $B \sim B + \Theta$ is exactly an $O(d, d; \mathbb{Z})$ duality of the DFT background (combined with `TDuality.thetaShift_isODD`, the real Θ -shift also preserves η : `thetaShiftR_preserves_eta`). Shifting by Θ and then by $-\Theta$ returns the original background (`genMetric_bshift_cancel`). The overall sign/transpose convention in the formula above was checked with `sympy` *before* the Lean statement was written (`symbolic check (sympy)`, not Tier A): of four candidate placements of $\pm\Theta$ and transposes, exactly this one holds; the Lean theorem is the resulting Tier A statement, not the `sympy` check itself.

4.3 Level matching and the section condition

Hull–Zwiebach’s level-matching condition, $L_0 - \bar{L}_0 = N - \bar{N} - p_a w^a = 0$ (eq. (1.3), l. 231), at equal oscillator number $N = \bar{N}$ reduces to $n \cdot w = 0$, i.e. the charge is η -null. `DFT.SectionCondition` proves this equivalence exactly (`levelMatching_iff`: `chargeNorm(n, w) = 0  $\iff$  n · w = 0`). A set of charges is a *section* if it is totally η -null (`IsSection`); Hull–Zwiebach’s description of the two extreme solutions of the strong section condition — “independent of $\tilde{x}$ ” (the ordinary supergravity fields) and “independent of x^a ” (their duals), ll. 339–342, and “totally null d -dimensional subspaces,” ll. 3930–3940 — correspond here to the pure-momentum frame and the pure-winding frame, each proved to be a section (`momentumFrame_isSection`, `windingFrame_isSection`). The full duality η maps the momentum frame onto the winding frame (`eta_`

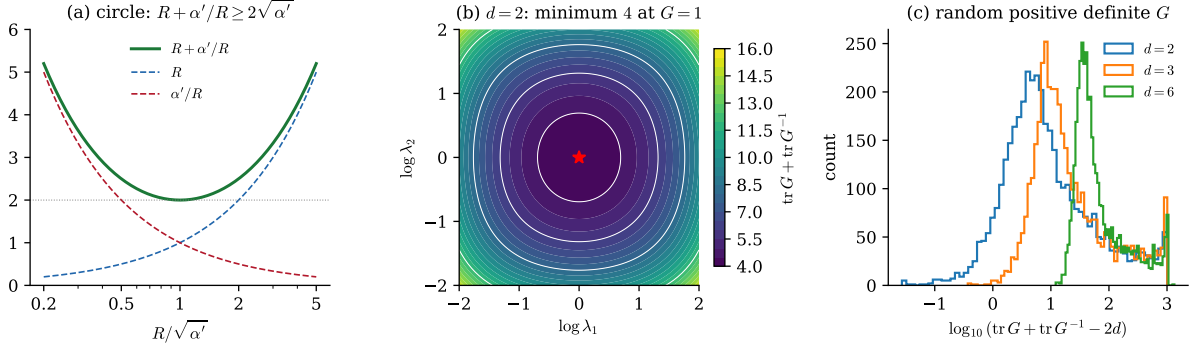


Figure 6: The dual-scale bound. (a) On a circle the effective scale $R + \alpha'/R$ is minimized at the self-dual radius. (b) For $d = 2$ and positive-definite G with eigenvalues $\lambda_{1,2}$, $\text{tr } G + \text{tr } G^{-1} = \sum_i (\lambda_i + \lambda_i^{-1})$ has its minimum $2d = 4$ at $G = 1$. (c) Excess over $2d$ for 4000 random positive definite matrices in each dimension: never negative, as the Lean theorem `dualScale_ge` guarantees. The plot illustrates the theorem; it proves nothing. *Computed figure; data: Lean dualScale_circle, circle_effective_scale_ge_two, dualScale_ge* (TraceBound.lean).

`momentum_to_winding`), and any $g \in O(d, d; \mathbb{Z})$ maps sections to sections (`isSection_image`) — the section condition is preserved along the whole duality orbit, not merely at the two frames exhibited by name.

Table 3: Double Field Theory layer: Lean declaration and epistemic tier.

Lean declaration	Tier / content
<code>etaR_genMetric_sq</code>	A — $(\eta\mathcal{H})^2 = 1$ for any invertible G (sharper than HZ)
<code>genMetric_symm</code>	A — $\mathcal{H}$ symmetric for G symm., B antisymm.
<code>tduality_inverts_metric</code>	A — $\eta\mathcal{H}(G, 0)\eta = \mathcal{H}(G^{-1}, 0)$
<code>massForm_covariant</code>	A — covariance under <i>any</i> g (no $O(d, d)$ hypothesis)
<code>massForm_circle</code>	A — $d = 1$ DFT mass form = $\frac{1}{2} \times \text{Stream 1's momentumMassSq}$
<code>genMetric_bshift</code>	A — $g_\Theta \mathcal{H}(G, B) g_\Theta^\top = \mathcal{H}(G, B + \Theta)$
Sign/transpose convention	symbolic check (sympy), not Tier A — sympy, checked before formalization
<code>thetaShiftR_preserves_eta</code> , <code>genMetric_bshift_cancel</code>	A — real Θ -shift $\in O(d, d)$; shift-and-unshift
<code>levelMatching_iff</code>	A — $L_0 = \bar{L}_0 \iff n \cdot w = 0$
<code>momentumFrame_isSection</code> , <code>windingFrame_isSection</code>	A — pure-momentum/-winding frames are sections
<code>eta_momentum_to_winding</code> , <code>isSection_image</code>	A — η swaps frames; $O(d, d; \mathbb{Z})$ preserves sections

5 The dual-scale trace bound

This research programme’s recurring proposal (Stream 1, `DualScaleM24Formalization.DualScale.Effective no_singularity`, and paper 7’s “The Effective Dual Scale and Singularity Resolution” section) is the one-dimensional statement that the effective scale probed by a string, $R_{\text{eff}}(R) = R + \alpha'/R$, is bounded below by $2\sqrt{\alpha'}$, attained at the self-dual radius. `DualScale.TraceBound` generalizes this to every flat torus metric G (a positive-definite $d \times d$ real matrix), defining the *dual scale* as the trace of the generalized metric at $B = 0$:

$$\mathcal{D}(G) := \text{tr } \mathcal{H}(G, 0) = \text{tr } G + \text{tr } G^{-1} \quad (\text{dualScale_eq}). \quad (12)$$

This is invariant under the full T-duality, $\mathcal{D}(G^{-1}) = \mathcal{D}(G)$ (`dualScale_inv`), and satisfies, for every positive-definite G ,

$$\mathcal{D}(G) \geq 2d \quad (\text{dualScale_ge}), \quad (13)$$

proved directly from $\text{tr}((G - 1)^\top G^{-1}(G - 1)) \geq 0$ (positive semi-definiteness of a conjugate of a positive-definite matrix), which expands to $\text{tr} G + \text{tr} G^{-1} - 2d \geq 0$ — *no eigenvalue decomposition of G is used*. The bound is attained at the self-dual metric $G = \mathbf{1}_d$, $\mathcal{D}(\mathbf{1}_d) = 2d$ (`dualScale_one`); **the Lean statement proves the lower bound and its attainment at $G = 1$, not that $G = 1$ is the unique minimizer** — uniqueness of the minimizing metric is not claimed or proved here, and this paper does not write “equality iff” for this theorem. At $d = 1$, $G = R^2$,

$$\mathcal{D}(R^2) = (R + R^{-1})^2 - 2 \quad (\text{dualScale_circle}), \quad (14)$$

and specializing `dualScale_ge` to $d = 1$ together with this identity gives exactly

$$R + R^{-1} \geq 2 \quad \text{for all } R > 0 \quad (\text{circle_effective_scale_ge_two}), \quad (15)$$

the real-number form of the one-dimensional dual-scale bound, now proved over $\mathbb{R}$ (using Mathlib) rather than only the N-instance available to the Mathlib-free Stream 1 corpus.

Physical reading (Tier C). At the self-dual point $G = \mathbf{1}_d$ the trace bound is saturated: reading $\mathcal{D}(G)$ as a proxy for an effective probed length scale, no $O(d, d; \mathbb{Z})$ -invariant combination of the torus radii can be driven below this value by any sequence of T-dualities. As in paper 7, we flag this as exactly the same kind of *kinematic* statement as the circle case: a bound on a specific algebraic quantity built from the metric, not a dynamical proof that any cosmological evolution of $G(t)$ must bounce through this regime, and not a proof of a minimum *physical* length in any operational sense. Any singularity-resolution or minimum-length reading of this bound remains Tier C, exactly as it was for the $d = 1$ statement in Stream 1 and paper 7.

Table 4: Dual-scale bound: Lean declaration and epistemic tier.

Lean declaration	Tier / content
<code>dualScale_eq</code>	A — $\mathcal{D}(G) = \text{tr } G + \text{tr } G^{-1}$
<code>dualScale_inv</code>	A — $\mathcal{D}(G^{-1}) = \mathcal{D}(G)$
<code>dualScale_ge</code>	A — $\mathcal{D}(G) \geq 2d$ for G positive definite
<code>dualScale_one</code>	A — bound attained at $G = \mathbf{1}_d$ (attainment only, not uniqueness)
<code>dualScale_circle</code>	A — $d = 1$: $\mathcal{D}(R^2) = (R + R^{-1})^2 - 2$
<code>circle_effective_scale_ge_two</code>	A — $R + R^{-1} \geq 2$, real, for all $R > 0$
Singularity resolution / minimum length	C — physical reading, not proved

6 Flux and tadpole cancellation on $K3 \times T^2$ and $K3 \times K3$

Dasgupta, Rajesh, and Sethi study M-theory and type IIB orientifold compactifications on $K3 \times T^2/K3 \times K3$ with G-flux (hep-th/9908088) [9]. For M-theory on $K3 \times K3$, “the anomaly $\chi/24 = 24 \dots$ must be cancelled by a combination of branes and G-flux” (§3.1, l. 640); the type IIB orientifold of $K3 \times T^2$ is the F-theory dual of that setup (ll. 1265ff), with tadpole condition $\frac{1}{2} \int (G/2\pi) \wedge (G/2\pi) + n = 24$ (ll. 1366–1370), and with no flux, “type I on $K3 \times T^2$ with 24 D5-branes” (l. 1275).

`Flux.Tadpole` computes Euler characteristics from Hodge-number tables via $\chi = \sum_{p,q} (-1)^{p+q} h^{p,q}$ (`eulerFromHodge`), reusing Stream 1’s certified $K3$ Hodge diamond (`StringTheoryFormalization.Frontier.HodgeNumbers.k3HodgeNumber`) rather than re-tabulating it: $\chi(T^2) = 0$ (`euler_T2`) and $\chi(K3) = 24$ (`euler_K3`). By Künneth (Tier L; not separately formalized here), $\chi(K3 \times K3) = \chi(K3)^2$ (`chiK3K3`), giving $\chi(K3 \times K3)/24 = 24$ exactly, an integer (`k3k3_anomaly`), matching DRS’s stated anomaly value. Separately, $\chi(K3) \cdot \chi(T^2) = 0$ (`k3t2_euler_zero`): $K3 \times T^2$ itself contributes no curvature tadpole, consistent with the flux/brane budget being the only source of the anomaly there. The D3/flux

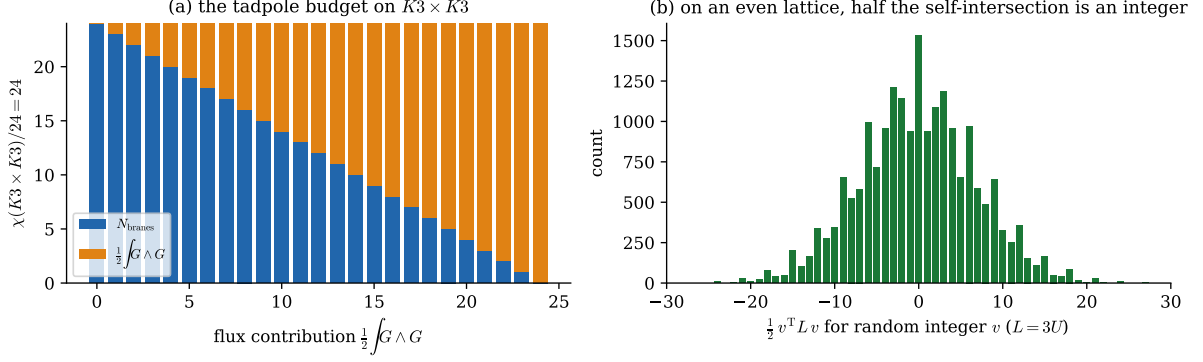


Figure 7: (a) Every way to split the tadpole budget $\chi(K3 \times K3)/24 = 576/24 = 24$ between space-filling branes and flux (`k3k3_anomaly`, `tadpole_budget`). (b) For random integer flux vectors on the even lattice $3U$, $\frac{1}{2}v^T L v$ is always an integer, the lattice fact behind flux quantization (`kronForm_even`, `flux_half_selfIntersection_integral`). *Computed figure; data: SVW ($\chi/24$ condition); DRS §2–4; Lean Flux/Tadpole.lean, Flux/Integrality.lean.*

budget arithmetic $\text{flux} + n = 24$, $\text{flux} \geq 0 \Rightarrow n \leq 24$, and $\text{flux} = 0 \Rightarrow n = 24$ (`tadpole_budget`) is a direct consequence of the stated tadpole condition; nothing about the *existence* of a flux configuration achieving any particular budget is claimed.

Why the flux contribution is an integer in the first place is a separate fact, proved in `Flux.Integrality`: DRS’s Dirac-quantization argument (ll. 230–232, “If $\chi/24 \in \mathbb{Z}$ then $[G/2\pi]$ is an integer cohomology class”) combined with Künneth’s identification $H^2(K3) \otimes H^2(K3) \subset H^4(K3 \times K3)$ (both Tier L) reduces the claim $\frac{1}{2} \int G \wedge G \in \mathbb{Z}$ to a statement about the Kronecker product of the $K3$ intersection form with itself. `Flux.Integrality` proves, for *any* two symmetric integer forms L_1, L_2 with L_1 even, that their Kronecker product $L_1 \otimes L_2$ is again an even form (`kronForm_even`), and hence that every self-intersection $x^T(L_1 \otimes L_2)x$ is exactly twice an integer (`flux_half_selfIntersection_integral`: $\exists k, x^T(L_1 \otimes L_2)x = 2k$). Since the $K3$ lattice’s summands U and $E_8(-1)$ are already proved even (Section 2), this instantiates to flux forms $G \in H^2(K3) \otimes H^2(K3)$. The Lean theorem is purely about the Kronecker product of two abstract integer forms; the identification of G with a flux configuration and of $x^T(L_1 \otimes L_2)x$ with $\int G \wedge G$ is the Künneth/Dirac-quantization input, Tier L. Gukov–Vafa–Witten’s superpotential for flux compactifications on Calabi–Yau fourfolds [10] is cited here only as context for why flux superpotentials of this kind matter physically on $K3 \times T^2$; no Lean module in this corpus formalizes any part of the GVW superpotential itself.

Table 5: Flux and tadpole layer: Lean declaration and epistemic tier.

Lean declaration	Tier / content
<code>euler_T2</code> , <code>euler_K3</code>	A — $\chi(T^2) = 0$, $\chi(K3) = 24$ from Hodge tables
<code>k3k3_anomaly</code>	A — $\chi(K3 \times K3)/24 = 24 \in \mathbb{Z}$ (Künneth is L)
<code>k3t2_euler_zero</code>	A — $\chi(K3)\chi(T^2) = 0$
<code>tadpole_budget</code>	A — arithmetic consequence of $\text{flux} + n = 24$
Tadpole condition, DRS eqs. (line refs above)	L — Dasgupta–Rajesh–Sethi
<code>kronForm_even</code> , <code>flux_half_selfIntersection_integral</code>	A — even $\otimes$ any = even; self-intersections are $2k$
Künneth identification with $\int G \wedge G$; Dirac quantization	L — DRS ll. 230–232; Künneth
GVW flux superpotential	L — context only, not formalized

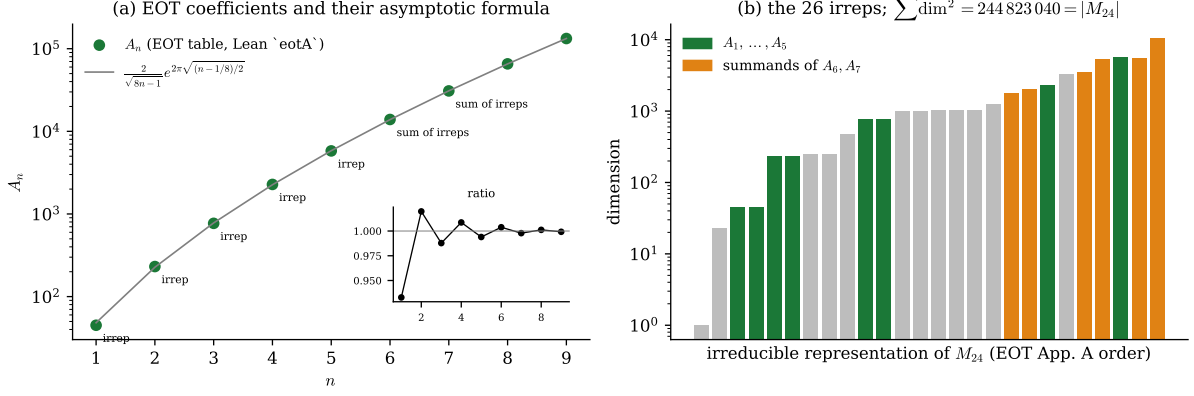


Figure 8: (a) The coefficients A_n of the K3 elliptic genus (EOT eq. 1.12, the Lean table `eotA`) against the asymptotic formula EOT eq. (1.13); the inset shows the ratio. $A_1, \dots, A_5$ are dimensions of irreducible representations of M_{24} ; A_6 and A_7 are sums of them. (b) The 26 irreducible dimensions of M_{24} from the Lean table `M24RepDim`; the sum of their squares equals the group order (`M24RepDim_sum_sq`), the check that caught an earlier wrong table. *Computed figure; data: EOT eqs. (1.12)–(1.15), ll. 200–232 (formula read from the PDF page, the .txt garbles it); App. A (A.3).*

7 Mathieu moonshine arithmetic

Eguchi, Ooguri, and Tachikawa (EOT) observed that the coefficients of the $\mathcal{N} = 4$ superconformal character decomposition of the K3 elliptic genus,

$$Z_{\text{ell}}(K3) = 20 \text{ch}_{1/4,0} - 2 \text{ch}_{1/4,1/2} + 2 \sum_{n \geq 1} A_n q^{n-1/8} \theta_1^2 / \eta^3 \quad (16)$$

(eq. (1.11), ll. 175–195), with $A_n = 45, 231, 770, 2277, 5796, 13915, \dots$ (eq. (1.12), l. 208), are *dimensions of irreducible representations* of the sporadic Mathieu group M_{24} for the first five values (ll. 227–233), a phenomenon subsequently proved in full by Gannon [12]. **Notational note:** papers 2 and 7 of this project’s own corpus use the symbol A_n for *twice* EOT’s coefficients (their $\mathcal{A}_n := 2A_n^{\text{EOT}}$, so that, e.g., $A_1 = 90 = 45 \oplus 45$); `Moonshine.EOT` uses EOT’s own, undoubled convention, `eotA` = (45, 231, 770, 2277, 5796, 13915, 30843, 65550, 132825) matching eq. (1.12) directly, and this paper follows `Moonshine.EOT`’s convention throughout this section to avoid re-introducing the ambiguity flagged in the revision brief. Also for the record, since an earlier revision of paper 7 stated it incorrectly: EOT’s eq. (A.3) (ll. 346–349) lists the irreducible dimensions 45, 231, 770, 990, 1035, $\dots$ as coming in complex-conjugate pairs, while 2277 is a *real* irreducible dimension, so the doubled multiplicity is $4554 = 2 \cdot 2277$, two copies of the same real representation, not $2277 \oplus \overline{2277}$.

`Moonshine.EOT` checks the EOT arithmetic against Stream 1’s own M_{24} irreducible dimension table, `StringTheory.StringDynamics.MathieuM24.M24RepDim` (itself guarded by the Burnside identity $\sum_i \dim(i)^2 = |M_{24}| = 244,823,040$ in `M24RepDim_sum_sq`, a Stream 1 theorem, cited here as the safeguard that this table is correct and not, as an earlier version of it briefly was, wrong by a duplicated and a missing entry). `first_five_are_irreps` proves that each of $A_1, \dots, A_5$ equals some entry of `M24RepDim` (witnesses at table indices 2, 4, 9, 19, 23 for 45, 231, 770, 2277, 5796 respectively); this is an equality against Stream 1’s table, so its physical content — that these table entries really are M_{24} ’s irreducible dimensions — rests on that table’s Tier L sourcing (EOT eq. (A.3)) via Stream 1, not re-derived here. `A6_decomposition` proves EOT’s eq. (1.14), $A_6 = 13915 = 3520 + 10395$ (table indices 21, 25), and `A7_decomposition` proves eq. (1.15), $A_7 = 30843 = 10395 + 5796 + 5544 + 5313 + 2024 + 1771$ (indices 25, 23, 24, 22, 18, 17); both by `rf1` on the now-corrected table. `A6_not_irrep` confirms that A_6 itself is *not* the dimension of any

single irreducible representation in the table, so a genuine decomposition (rather than a single witness, as for $A_1, \dots, A_5$) is required for it. Throughout, that the elliptic genus decomposition in fact carries an honest M_{24} *action* (rather than this arithmetic coincidence) is the moonshine phenomenon itself — EOT’s observation, proved by Gannon [12] — and is Tier L, not addressed by this arithmetic module.

Table 6: Moonshine layer: Lean declaration and epistemic tier.

Lean declaration	Tier / content
eotA	def — EOT eq. (1.12) coefficients, undoubled convention
first_five_are_irreps	A — $A_1, \dots, A_5$ each equal an M24RepDim table entry
A6_decomposition, A7_decomposition	A — EOT eqs. (1.14)/(1.15) sums vs. table
A6_not_irrep	A — A_6 is not a single table entry
M24RepDim_sum_sq	A — Stream 1, Burnside guard $\sum \dim^2 = M_{24} $
M_{24} irreducible dimensions (eq. A.3); M_{24} action on the genus	L — EOT; Gannon

8 What is, and is not, proved

Scope of Lean 4 verification

Is proved (Tier A, kernel-checked, 99 theorems, 0 sorry, standard axioms only):

exact integer/rational identities about explicit matrices — unimodularity and positive-definiteness of E_8 and $E_8(-1)$; evenness, unimodularity, and a signature witness for U ; signature and rank arithmetic for the $K3$, Mukai, and $K3 \times T^2$ lattices, cross-checked against an independently tabulated Hodge signature; symmetry, self-pairing, and evenness of the Mukai pairing; that (-2) -vector reflections are involutive isometries, instantiated to the simple roots of $E_8(-1)$; that specific matrix families (Θ -shifts under an antisymmetry hypothesis, basis changes under a unimodularity hypothesis, factorized dualities, η itself) lie in $O(d, d; \mathbb{Z})$ for *every* d , and that $O(d, d; \mathbb{Z})$ is closed, every element invertible, and charge-norm- and (mass², level)-preserving under conjugation; the Hull–Zwiebach generalized metric’s defining constraint $(\eta\mathcal{H})^2 = 1$ (proved under a weaker hypothesis than the source states), its symmetry, its B -shift covariance, and that T-duality is metric inversion; level matching as η -nullity and the section-condition frames; the d -dimensional dual-scale bound $\text{tr } G + \text{tr } G^{-1} \geq 2d$ and its attainment at $G = 1$; flux-form evenness and the resulting tadpole integer arithmetic on $K3 \times T^2/K3 \times K3$; and the EOT elliptic-genus coefficient decompositions against a Burnside-guarded M_{24} table.

Is not proved: that any of these matrices, groups, or lattices *is* a string-theory charge lattice, duality group, or background field — every such identification is cited from the literature (Tier L, pinned to file and line throughout Sections 2 to 4, 6 and 7) or is this project’s own physical reading (Tier C, Section 5’s singularity/minimum-length narrative); that the exhibited $O(d, d; \mathbb{Z})$ generators generate the full group (Tier L, GPR); Sylvester’s law of inertia, the mod-8 theorem for even unimodular indefinite lattices, or the classification of $K3$ moduli space (all Tier L, none re-derived); the converse directions of several “if” statements (`thetaShift_isODD`, `basisChange_comm_thetaShift`) — one of which was checked only symbolically (sympy) and is explicitly flagged as not Tier A; that $G = 1$ is the *unique* minimizer of the dual-scale bound; invariance of the full string partition function under T-duality (only the discrete charge spectrum is matched); the existence of any actual flux configuration achieving a stated tadpole budget; and that the elliptic-genus coefficients carry a genuine M_{24} *action* rather than merely matching its representation dimensions (Gannon’s theorem, Tier L). No claim in this paper is that any part of string theory itself has been formalized, proved, or certified.

Declaration dependency graph (840 declarations, 1110 edges); node size = number of dependents

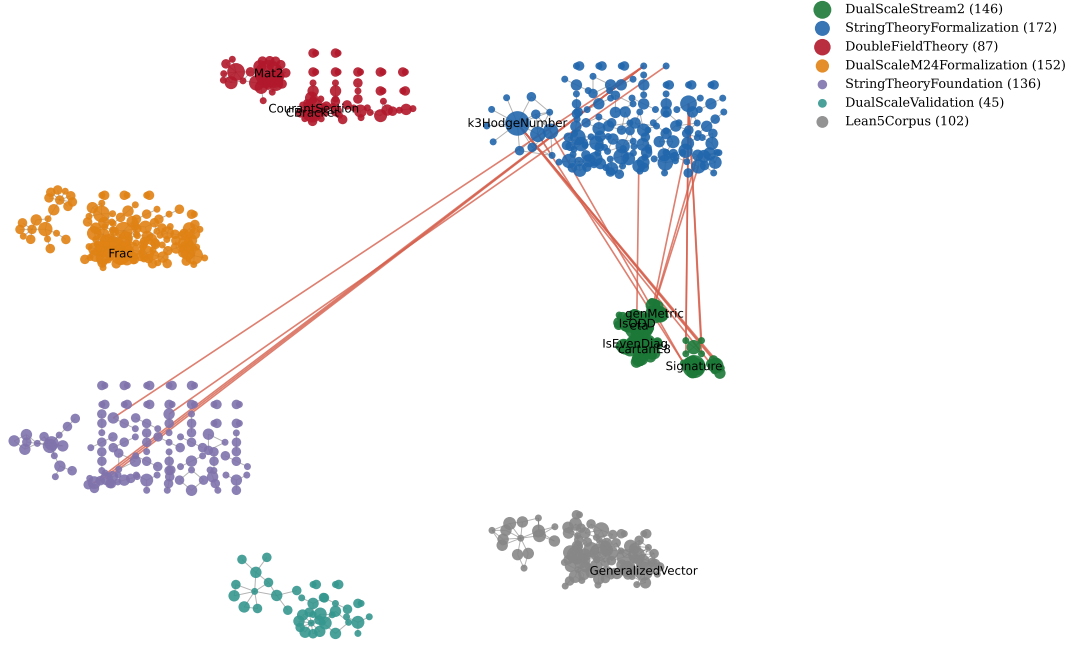


Figure 9: Kernel-level dependency graph of the declarations written in the source (isolated declarations omitted), one cluster per library (spring layout inside each cluster). Node size grows with the number of declarations that use it; the largest hubs are labelled. Red edges cross library boundaries: they are the bridges listed in the atlas chapter. *Computed figure; data: `tools/lean_depgraph.lean` (`.leancache/depgraph.jsonl`); layout: `networkx` Kamada-Kawai per connected component, one cluster per library.*

9 Methods: a tiered agent pipeline

9.1 Pipeline

Every statement in `DualScaleStream2` was designed by an orchestrating model with each Tier L source pinned to a specific file and line range in `papers/foundations/`, so that a script can `grep` the citation rather than trust it from memory. Proofs were then searched in cost-ascending tiers, escalating a goal only after the cheaper tier failed under the kernel gate: a local 7B prover model (DeepSeek-Prover-V2, quantized Q8_0, run on a single T4 GPU) first; then Claude Haiku for mechanical residuals; then Claude Sonnet for goals needing an actual proof idea (block-matrix identities involving inverses, positive-definiteness arguments, the diagonal plus strict-upper-triangular decomposition used in `even_quadratic_form_of_even_diag`). An orchestrating model (Opus) selected work packages, reviewed each Lean *statement* against its cited source before any proof attempt (the one step no kernel gate can check — whether the formal statement says what the paper claims it says), assigned the A/L/C tier, and committed. The accept gate at every stage was strictly the Lean kernel: `lake build DualScaleStream2` green, a zero-sorry audit, and `tools/axiom_audit.py`'s `#print axioms` showing only the three standard axioms; no `native_decide` is used anywhere in this library. Producer and verifier were kept separate throughout: no agent's own report of success was counted without an independent recompilation.

9.2 Measured results

Full per-package tables are in `docs/STREAM2_WORKFLOW.md` §6; we summarize the pattern here rather than re-deriving a single global percentage (the package-level goal counts there — 26, 23, 10, and 6 goals for the lattice/T-duality pilot, the P2.3/P2.4-core/P2.5/P2.6 batch, the P2.2 factorized-duality batch, and the dual-scale bound, respectively, summing to 65 of the project’s 99 final theorems — are reported *per package*, not as a global tier table; the remaining goals were closed in later rounds, described qualitatively rather than re-tabulated, on the mirror identity, section-condition, $SL(2, \mathbb{Z})$ -product, B -shift, spectrum-equivalence, reflection, and flux-integrality modules). The qualitative pattern was consistent and, in the lattice pilot, sharp: the local prover closed essentially every *concrete* goal (fixed matrices, numerals, the 64-entry E_8 inverse check by `fin_cases`, at a median latency of 4.4 s per attempt after a roughly 170 s one-time cold load) and **zero** of the eight general- n /general- d symbolic goals in that pilot. Haiku closed most short symbolic algebra with an orchestrator-written proof sketch; Sonnet was needed only where a real mathematical idea was required. The Sonnet share varied sharply by package rather than sitting at one corpus-wide rate: in the lattice pilot, Sonnet closed 3 of 26 goals (11.5%); in the P2.3/P2.4-core/P2.5/P2.6 batch, 8 of 23; in the P2.2 factorized-duality batch, 7 of 10; in the dual-scale bound, 2 of 6. A second local model, Goedel-Prover-V2-8B, closed zero goals in the pilot: with its thinking mode on it exhausted an 8192-token, 615-second budget without emitting an answer, and with thinking off it drifted off target and emitted **sorry** scaffolds; this was recorded as a configuration failure at this hardware’s speed, not a capability verdict, and the model was dropped from routing.

9.3 Honest failures

Two Haiku agent reports misstated their own compile status and were caught only because the orchestrator recompiled every report before counting it (producer $\neq$ verifier, enforced, not merely stated): one claimed five clean **sorry** placeholders while actually leaving eight compile errors in **GeneralizedMetric**; another claimed “no compilation errors” in **Factorized** while three proofs failed to parse. In the corpus’s final round, a Haiku report again claimed “0 errors” on the section-condition goals while one proof in fact failed to parse due to an operator-precedence mistake, repaired by the orchestrator before the gate was re-run. Separately, `lake env lean` (used by the single-file prover loop) does not apply `lakefile.lean`’s `maxHeartbeats` option, so agents were, for a time, compiling under a stricter timeout than the actual build uses; one Haiku agent abandoned the E_8 LDL^T goal on a heartbeat timeout that the real build budget does not hit, a lost-time bug rather than a soundness one (a stricter gate can only reject valid proofs, never accept invalid ones). The route chosen for a goal mattered as much as the tier assigned to it: `cartanE8_posDef` failed under Haiku via the literal 64-entry $L \cdot D \cdot L^T$ matrix product, and was closed by Sonnet only after the orchestrator reformulated it as the sympy-verified sum-of-squares identity checked directly by `ring`. Finally, several block-matrix sign/transpose conventions — the B -shift covariance law, the mirror-symmetry matrix identity, the $O(d, d)$ pairing on J — were checked with sympy before the corresponding Lean statement was written; this project treats such a check as a useful pre-screen against a wasted proof attempt, and, per Section 1.3, never as a substitute for the kernel proof itself.

10 Conclusion

DualScaleStream2 kernel-certifies (Tier A) the algebraic skeleton of $K3 \times T^2$ T-duality and Double Field Theory: the positive-definiteness and unimodularity of the lattices that build $H^2(K3, \mathbb{Z})$, the signature bookkeeping from $K3$ up through the Mukai and $K3 \times T^2$ charge lattices, the $O(d, d; \mathbb{Z})$ generators and their charge- and spectrum-invariance for every torus dimension d , the Hull–Zwiebach generalized metric and its B -shift and section-condition structure,

a genuine real-analytic d -dimensional generalization of this programme’s dual-scale bound, flux-integrality and tadpole arithmetic on $K3 \times T^2$ and $K3 \times K3$, and the EOT elliptic-genus coefficient arithmetic against a Burnside-guarded M_{24} table. None of this proves that these objects constitute a string theory: the identification of each formal object with its physical counterpart is drawn from the cited literature or stated as this project’s own conjecture, tier by tier, throughout. We consider the value of this exercise to lie precisely in that separation — and in the pipeline’s own measured failures (a symbolic goal the local prover cannot touch, an agent report that is simply false, a build-configuration mismatch that silently over-constrains a proof search) being caught by a kernel gate and a producer-independent verifier rather than by trust.

A Reproducibility

A.1 Build

From the repository root:

```
lake build DualScaleStream2
```

completes in **3670 jobs** with no errors and zero `sorry` warnings, on toolchain `leanprover/lean4:v4.33.1` with Mathlib v4.33.1. The library comprises **19 modules** across the `Lattice`, `TDuality`, `DFT`, `DualScale`, `Flux`, and `Moonshine` namespaces (listed in Sections 2 to 7). Most of the 3670 jobs are Mathlib’s own dependency graph, not this library’s proof content; the job count is a build-system measurement, not a count of theorems.

A.2 Axiom audit

```
python3 tools/axiom_audit.py DualScaleStream2
```

reports **99 theorems**, **0 failing**, with every declaration’s axiom footprint limited to `propext`, `Classical.choice`, and `Quot.sound` — Lean’s three standard axioms, no custom axiom, and no `sorryAx` or `native_decide` anywhere in the library. Stream 1 was re-gated in the same session for comparison: `lake build` 3353 jobs across its six libraries, **326 theorems**, 0 failing (`DualScaleValidation` 23, `Lean5Corpus` 53, `DoubleFieldTheory` 44, `StringTheoryFoundation` 56, `DualScaleM24Formalization` 61, `StringTheoryFormalization` 89).

A.3 Statement lock

```
python3 tools/statement_lock.py -check
```

verifies every theorem statement (up to its top-level `:=`) and every definition body against a hash recorded in `docs/statement_lock.json` at review time, so that replacing a `sorry` with a proof cannot silently coincide with weakening the statement that was reviewed.

A.4 Roadmap coverage

`docs/STREAM2_WORKFLOW.md` §5.1 records **18 of 18** planned roadmap items (phases P2.1–P2.6) as done, each item meaning every theorem for it is committed, `sorry`-free, and axiom-audited. As the workflow document itself stresses, this is 100% *of this plan*, counting items against a checklist this project wrote for itself — not a fraction of “string theory,” and not a claim of completeness for any of the physical theories discussed in Section 1.

A.5 Scope of this appendix

No number in this appendix was taken from memory: the job counts, theorem counts, and axiom footprints above are the output of the commands shown, re-run at the commit recorded in this paper’s version history, and the phase/module correspondence matches `docs/STREAM2_WORKFLOW.md` §5.1 and §6 and `docs/PAPER_REVISION_BRIEF_2026_09_17.md` §1, which a reader can re-run rather than trust.

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